



Evolution of defects during the degradation of metal halide perovskite solar cells under reverse bias and illumination

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The efficiency and stability of perovskite solar cells are essentially determined by defects in the perovskite layer, yet their chemical nature and linking with the degradation mechanism of devices remain unclear. Here we uncover where degradation occurs and the underlying mechanisms and defects involved in the performance degradation of p-i-n perovskite solar cells under illumination or reverse bias. Light-induced degradation starts with the generation of iodide interstitials at the interfacial region between the perovskite and both charge transport layers. While we observe trap annihilation of two types of iodide defect at the anode side, we find negatively charged iodide interstitials near the cathode side, which we show to be more detrimental to the solar cell efficiency. The reverse-bias degradation is initialized by the interaction between iodide interstitials and injected holes at the interface between the electron transport layer and the perovskite. Introducing a hole-blocking layer between the layers suppresses this interaction, improving the reverse-bias stability.

Eliminating deleterious defects in semiconductors or passivating them during the fabrication process of solar cells has become one of the most fundamental tasks for the solar cell society^{1–4}. This scenario is also prevailing in the metal halide perovskite solar cell community, which has witnessed a rapid increase of the power conversion efficiency (PCE) of perovskite solar cells from 3.8% to 25.5% with overwhelming reported progress on defect passivation strategies^{5–8}. Many reported perovskite solar cells can now pass a 1,000 h operation stability test^{9,10}. Any further improvement of the efficiency or stability of perovskite solar cells towards their Shockley–Queisser limitations have to rely on deeper understandings on the nature of defects in perovskite to squeeze out all non-radiative charge recombination paths by eliminating or passivating them. The degradation of solar cells under bias or illumination is strongly related to defect evolution^{11–14}. However, it is still a challenge to experimentally identify the chemical nature of defects in perovskites.

Defects in perovskites have been intensively studied in recent years, but there is still no consensus on the defect chemical nature, their distribution and their evolution during degradation. Several calculations gave controversial results—that is, different defects, including halide interstitials (I_i^- and I_i^+), metal vacancies (V_{Pb}) or antisite (I_{MA})—that cause deep traps in methylammonium lead triiodide (MAPbI₃)^{15–19}. Recent experimental attempts to identify the chemical nature of defects in perovskites have implied the possible origins of the negatively charged iodide interstitial (I_i^-), MA vacancy (V_{MA}) and MA interstitial (MA_i^+) for deep traps in MAPbI₃ by deep-level transient spectroscopy characterizations²⁰; I_i^- as the dominant shallow anion defects in formamidinium lead triiodide (FAPbI₃) with positron annihilation spectroscopy measurement²¹ or FA₁ antisite as the dominant surface deep trap defects for FAMA perovskite²². However, there is no direct experimental approach to identify the chemical nature of defects in bulk and at the surface of

perovskites, making it difficult to understand the impact of defects on perovskite stabilities^{4,23}. It was recently demonstrated that the drive-level capacitance profiling (DLCP) technique was a feasible method to profile both the spatial and energetic distribution of traps in perovskite solar cells^{24,25}. This technique thus opens a new avenue to investigating the physical or chemical properties of traps in perovskites via the direct observation of the changes of traps in space upon environmental conditions.

In this Article, we combine the DLCP technique with electrical poling to determine the charge states and eventually the chemical nature of mobile defects in perovskites. Poling can manipulate the charge trap distribution in perovskite solar cells because all the ions that cause defects in perovskites are charged. This method allows us to study the degradation mechanism of perovskite solar cells under reverse biases, mimicking the shaded cells in a module. It also reveals the influence of the defects on the degradation of perovskite solar cells under illumination, pointing out most beginning degradation locations in perovskite solar cells and the heterogeneous defect generation and annihilation behaviours at the two interface regions of perovskite solar cells. The results highlight an important direction for the further enhancement of the photostability of perovskite solar cells.

Chemical nature of defects

We first studied MAPbI₃ thin single-crystal solar cells to find out the chemical nature of defects in MAPbI₃ crystals. The MAPbI₃ thin single crystal was synthesized using the space-confined lateral crystal-growing method with a thickness of ~35 μm (ref. ²⁶) and then converted to a solar cell with the device structure of indium tin oxide (ITO)/poly[bis(4-phenyl)(2,4,6-trimethylphenyl)amine] (PTAA)/MAPbI₃/fullerene (C₆₀)/bathocuproine (BCP)/copper (Cu), as schematically shown in Fig. 1a. We used the thermal admittance spectroscopy (TAS) method to characterize the energetic

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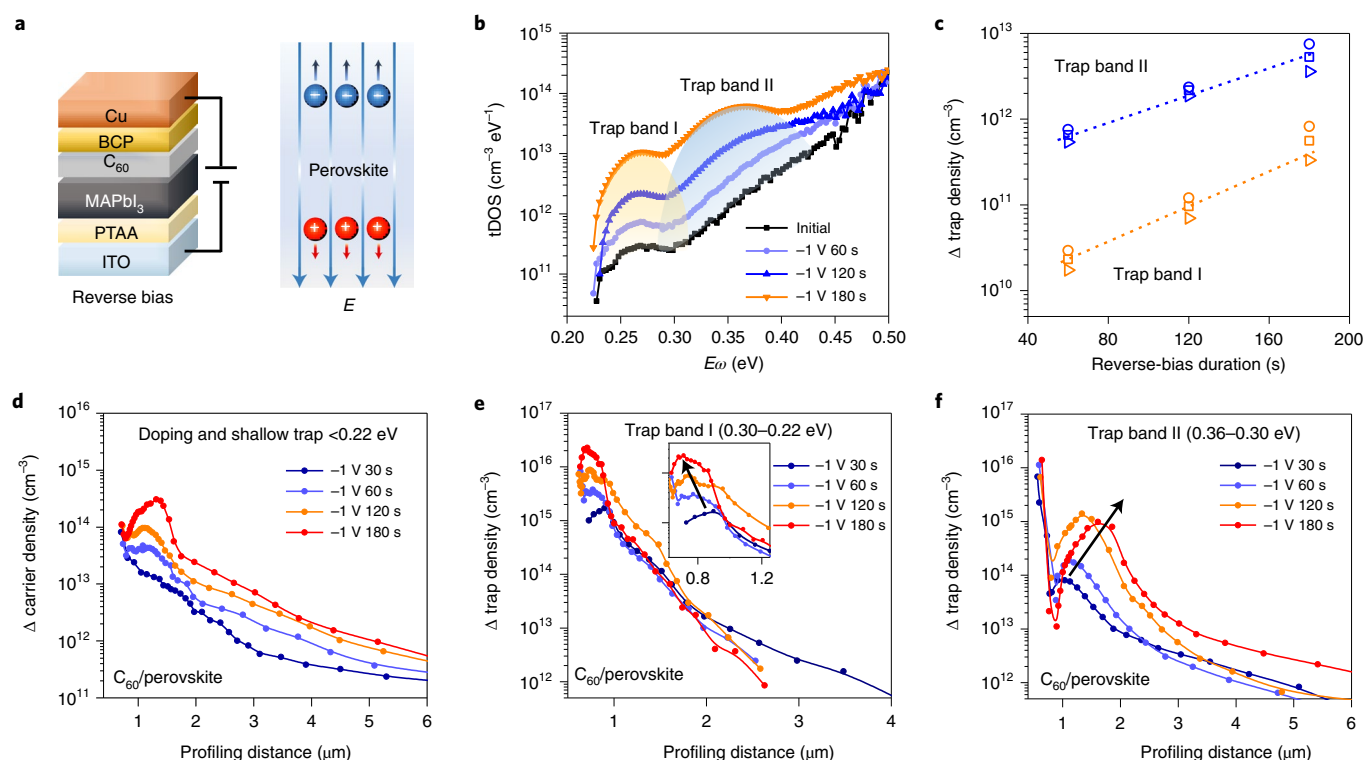


Fig. 1 | Degradation of thin single-crystal perovskite solar cell under reverse bias. **a**, Schematic diagrams of the device structure of a MAPbI₃ thin single-crystal solar cell under reverse bias (left) and the movement of charged ions in perovskite under electric field E (right). Positively and negatively charged ions are denoted by red and blue balls, respectively. The longer blue arrows indicate the direction of the electric field. **b**, tDOS spectra of a MAPbI₃ thin single-crystal solar cell before and after applying a reverse bias of -1 V with different durations. The orange and blue patterns denote the difference in tDOS before and after applying the reverse biases by performing Gaussian fitting to trap bands I and II, respectively. **c**, Δ trap density of trap bands I and II in MAPbI₃ thin single-crystal solar cells with different reverse-bias durations. Δ trap densities were calculated from the integrated trap densities after Gaussian fitting for results from three different devices, which were indicated by circle, square and triangle symbols, respectively. The dashed lines are guides to the eye for denoting the trend of Δ trap density with the increase of reverse-bias duration. **d–f**, Spatial distributions of the change in doping and shallow trap density with trap depth <0.22 eV (**d**), trap band I with the trap depth from 0.22 to 0.30 eV (**e**) and trap band II with the trap depth from 0.30 to 0.36 eV (**f**) in the MAPbI₃ thin single crystal after applying the reverse bias measured by DLCP. The inset in (**e**) displays the zoomed-in trap density change near the C₆₀/perovskite interface. The curves are guides to the eye. The black arrows denote the direction of the movement of the reverse bias-induced trap peaks with the extension of the bias duration. The applied bias was -1 V for all.

distribution of charge traps. Different trap bands in the MAPbI₃ single crystal, which are labelled as trap bands I and II with ascending trap depths, are shown in Supplementary Fig. 1. The measured energy depths for trap bands I and II are centred around 0.27 eV and 0.36 eV, respectively. The TAS-measured trap peaks were confirmed to be dominantly caused by the defects in the perovskite layer, rather than the charge transport layers, by comparing the devices with and without charge transport layers (Supplementary Note 1 and Supplementary Fig. 2)^{27,28}. The variations of the trap density of states (tDOS) of the MAPbI₃ thin single-crystal solar cell after biasing under a reverse bias of -1 V for different durations are shown in Fig. 1b, which shows the trap densities of both trap bands I and II gradually increased with bias duration. We calculated the reverse bias-induced trap densities (Δ trap density) in trap bands I and II by integrating the tDOS with respect to the energy depth for each trap band of three MAPbI₃ thin single-crystal solar cells (Supplementary Fig. 3a,c). The increment of the trap density of trap band II is almost tenfold to that of trap band I when the reverse-bias duration changed from 60 s to 180 s (Fig. 1c), indicating that the change in trap band II dominates the degradation of the MAPbI₃ thin single crystal under reverse bias.

To determine whether the defects for trap bands I and II are positively or negatively charged, we tracked the variation and

redistribution of defects under reverse bias in MAPbI₃ single-crystal solar cells. All perovskite solar cells in this work have been optimized to eliminate current and capacitance hysteresis before device stability testing (Supplementary Figs. 4 and 5). By varying the a.c. frequency (ω) of DLCP measurement, we separately measured the change of doping or trap distributions at the demarcation energy depth (E_{ω}) of 0.22 eV, 0.30 eV and 0.36 eV (Fig. 1d–f), from the variation of the total carrier densities under bias, which are shown in Supplementary Fig. 6. The change in doping and shallow traps with energy depths smaller than 0.22 eV ($\Delta n_{Et<0.22}$, where n_{Et} is trap density at the energy depth of E_t) in the MAPbI₃ thin single crystal when it was biased at -1 V for different durations is shown in Fig. 1d. Doping or charge traps were generated in the perovskite close to the C₆₀/MAPbI₃ interface, particularly with longer reverse-bias duration. The spatial distribution for the variation of trap band I (0.22 – 0.30 eV) and II (0.30 – 0.36 eV) in the MAPbI₃ solar cell under reverse bias is shown in Fig. 1e,f, respectively. Interestingly, both the trap densities of trap bands I and II increased in MAPbI₃ close to the C₆₀/MAPbI₃ interface after applying the reverse bias, and additional trap peaks induced by reverse bias appeared at the interface region. The emerged peak of trap band I shifted further towards the C₆₀/MAPbI₃ interface with a longer biasing (Fig. 1e). In contrast, the peak of the reverse bias-induced trap band II gradually moves

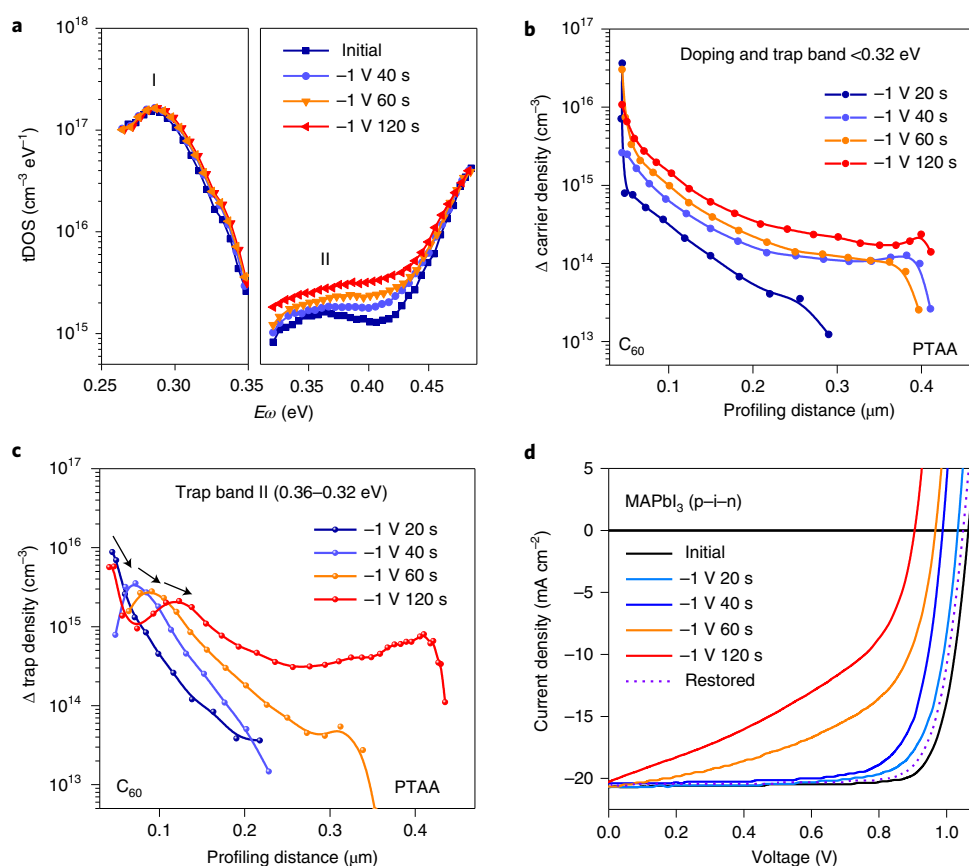


Fig. 2 | Degradation of polycrystalline thin film perovskite solar cells under reverse bias. **a**, tDOS spectra of a MAPbI₃ polycrystalline thin film solar cell before and after applying a reverse bias of -1 V with different durations. Trap bands I and II are indicated by Roman numerals I and II above the curves. **b,c**, Spatial distributions of the change in doping and trap density with trap depth <0.32 eV (**b**) and trap band II with the trap depth from 0.32 to 0.36 eV (**c**) in the MAPbI₃ film after applying the reverse bias of -1 V measured by DLCP. The note C₆₀ or PTAA indicates the location that is close to the C₆₀ or PTAA layer of the device. The curves are guides to the eye. The black arrows denote the direction of the movement of the reverse bias-induced trap peaks with the extension of the bias duration. **d**, J-V curves of the MAPbI₃ polycrystalline thin film solar cell before and after applying a reverse bias of -1 V with different durations. The dashed lines show the J-V curves of the restored devices after the revocation of the reverse bias and recovery in the dark for 2 h at room temperature.

towards the interior of the single crystal (Fig. 1f), accompanied by the broadening of the peak. Thus, defects for trap band I in MAPbI₃ are negatively charged, whereas trap band II is caused by positively charged defects. Similar results were observed in multiple MAPbI₃ thin single-crystal devices (Supplementary Fig. 3b,d). To rule out the possible influence of ion migration caused by the d.c. bias scanning during the DLCP measurement, we quickly cooled the samples to a low temperature of 260 K under the bias, and then we performed the DLCP measurements at low temperature. A similar trend was obtained for the movement of all trap bands (Supplementary Fig. 7), verifying that these changes are indeed caused by the larger reverse bias rather than the d.c. bias during measurement.

We then conducted the same measurements in MAPbI₃ polycrystalline solar cells. The tDOS result exhibited comparable trap depths for trap bands I and II in the MAPbI₃ polycrystalline thin film solar cells (Supplementary Fig. 8), which were around 0.29 eV and 0.36 eV, respectively. In contrast to the single-crystal solar cells, the trap density of trap band I in the polycrystalline solar cell is conspicuously higher than that of trap band II. We infer that trap band I is related to the defects that mainly locate at grain boundaries or amorphous regions of MAPbI₃ polycrystalline thin films. After applying a reverse bias of -1 V to the MAPbI₃ polycrystalline thin film solar cell, the trap density of trap band II clearly increased, whereas that of trap band I was almost unchanged (Fig. 2a and Supplementary Fig. 3e). This might be

explained by the much higher initial trap density of trap band I compared with that of trap band II in the polycrystalline film solar cells so that its variation can be shadowed by the background. DLCP results on the MAPbI₃ polycrystalline thin film solar cell after applying the reverse bias, which were derived from the total carrier density distributions measured at E_ω = 0.32 eV and 0.36 eV (Supplementary Fig. 9), are shown in Fig. 2b,c. Akin to the results obtained in the single-crystal solar cell, the doping and shallow trap densities (Δn_{E<0.32}) started to increase at the C₆₀/MAPbI₃ interface after applying the reverse bias (Fig. 2b). Meanwhile, a shoulder peak of trap band II appeared at the C₆₀/MAPbI₃ interface, which gradually shifted towards the interior of the film with the extension of the reverse-bias duration (Fig. 2c and Supplementary Fig. 3f).

Knowing the charge states greatly narrows down the possibility of the defects for trap bands I and II in MAPbI₃, which should be associated with the defects dominated by either iodine or MA, as lead defects are unlikely to move in MAPbI₃ due to its high ion-migration activation energy²⁹. In addition, antisite defects usually have higher formation energies compared with vacancies and interstitials as an antisite involves the formation of both vacancies and interstitials from different species⁴. We therefore tentatively assign trap band I to negatively charged I_i⁻ or V_{MA}⁻. And trap band II may be positively charged iodide interstitial (I_i⁺), iodide vacancy (V_I⁺) or MA_i⁺.

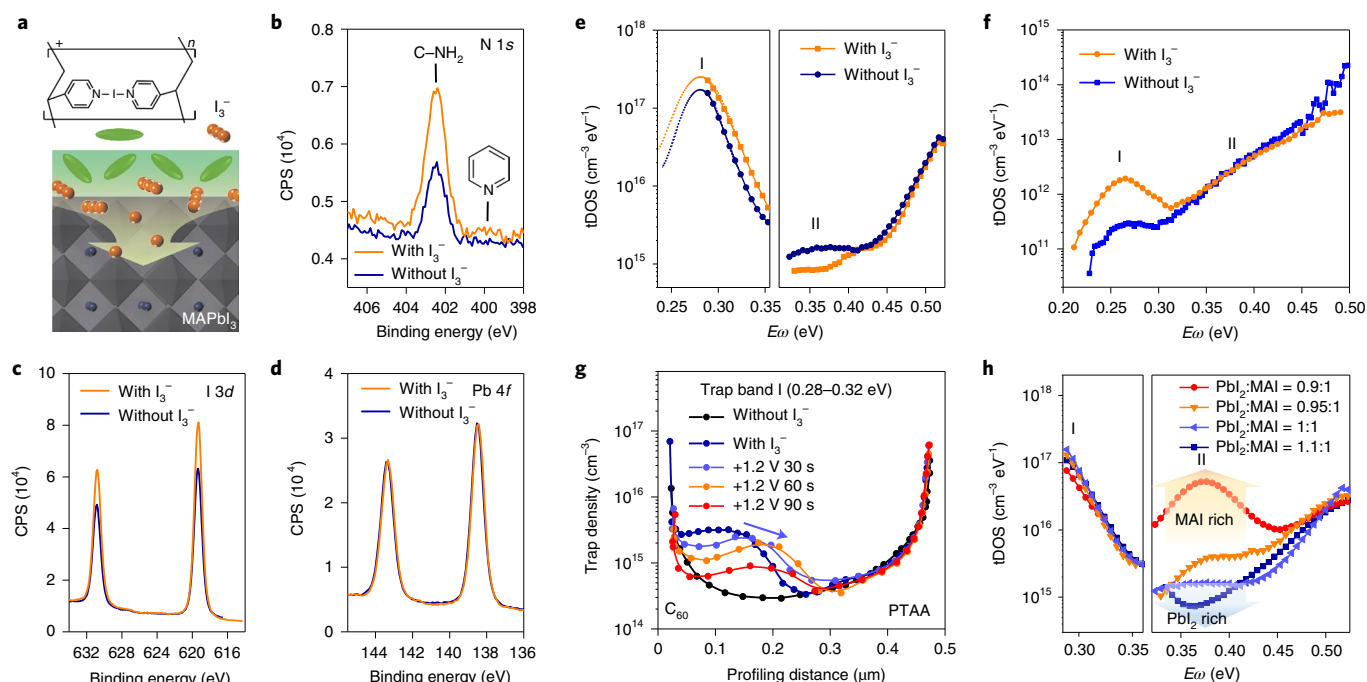


Fig. 3 | Identifying the chemical nature of traps in iodide perovskites. **a**, Schematic diagram of incorporating I_3^- ions on top of MAPbI_3 . The reaction between P4VP and I_2 in an isopropyl alcohol solution results in the formation of bis(pyridine)iodine cations (indicated by green ellipsoids) and I_3^- (indicated by orange balls). **b–d**, XPS spectra (counts per second (CPS) vs. binding energy) of N 1s (**b**), I 3d (**c**) and Pb 4f (**d**) of a MAPbI_3 polycrystalline thin film before and after the incorporation of I_3^- on top. The peaks at binding energies of 399.4 eV and 402.4 eV correspond to N in pyridine and C–NH₂ groups, respectively. Those at binding energies of 619.3 eV and 630.8 eV correspond to I^- , and binding energies of 138.5 eV and 143.3 eV correspond to Pb^{2+} in perovskite. **e,f**, tDOS spectra of the MAPbI_3 polycrystalline thin film (**e**) and thin single-crystal (**f**) solar cells before and after the incorporation of I_3^- on top of MAPbI_3 . Trap bands I and II are indicated by Roman numerals I and II above the curves. **g**, Spatial distributions of trap band I with the trap depth from 0.28 eV to 0.32 eV in the MAPbI_3 polycrystalline thin film solar cell before the incorporation of I_3^- , and after the incorporation of I_3^- with different durations under a forward bias of +1.2 V measured by DLCP. The note C_{60} or PTAA indicates the location that is close to the C_{60} or PTAA layer of the device. The curves are guides to the eye. The violet arrow denotes the direction of the movement of the forward bias-induced trap peaks with the extension of the bias duration. **h**, tDOS spectra of MAPbI_3 polycrystalline thin film solar cells with varying PbI_2 -to-MAI ratios of the precursors. The shaded yellow or blue arrows indicate the increase or decrease of trap band II in MAI-rich or PbI_2 -rich synthetic conditions for MAPbI_3 .

To further narrow down the possible origin of trap bands I and II, we intentionally introduced I_3^- ions by spin coating an I_2 -contained poly(4-vinylpyridine) (P4VP) solution on top of the MAPbI_3 polycrystalline thin films (Fig. 3a) and then detected the corresponding changes in trap states. The reaction between pyridine and I_2 would form charge transfer complexes that contain bis(pyridine)iodine cations and I_3^- (Supplementary Fig. 11a)³⁰. This is confirmed by Fourier-transform infrared measurement, which showed the signal of bis(pyridine)iodine cations at the wavenumber of 1,600 cm^{-1} (Supplementary Fig. 11b)³⁰. X-ray photoelectron spectra (XPS) further indicate that the atomic ratio of I^- to Pb^{2+} on the film surface increased from 2.31 to 2.91 after treatment (Fig. 3b–d and Supplementary Fig. 12). In addition, the atomic ratio of N in C–NH₂ (with a binding energy of 402.4 eV)³¹ to Pb^{2+} of the MAPbI_3 film increased from 0.63 to 0.93 as well. This implies that both I^- and MA^+ concentrations increased at the surface region of the MAPbI_3 polycrystalline thin film after the treatment. The relatively more MA^+ on the surface of P4VP– I_3^- treated films compared with that of untreated films should be due to the suppressing of MA^+ loss from the surface of P4VP– I_3^- treated MAPbI_3 during annealing or under high-vacuum conditions³². The slightly enhanced XPS intensity around binding energy of 399.4 eV was induced by the N in the pyridine group of the P4VP solution (Fig. 3b)³³. As shown in Fig. 3e, the density of trap band I, which is negatively charged, increased whereas that of trap band II, which is positively charged, slightly decreased after adding I_3^- . We did the same surface treatment on a

MAPbI_3 thin single crystal and found only the density of trap band I increased after treatment (Fig. 3f). The enrichment of MA^+ at the surface region of MAPbI_3 after adding I_3^- indicates a presumable reduction of V_{MA}^- or increment of MA_i^+ in MAPbI_3 , which does not follow the change of the trap bands I and II in MAPbI_3 after adding I_3^- , thus ruling out V_{MA}^- and MA_i^+ as the origin for trap bands I and II, respectively. These results demonstrate that trap band I of MAPbI_3 should be caused by I_i^- , whereas trap band II may arise from I_i^+ or V_i^+ .

DLCP measurements on MAPbI_3 thin film solar cells with the addition of I_3^- and under forward bias further verify that trap band I in MAPbI_3 is caused by I_i^- . The trap density of band I (0.28–0.32 eV) at near interface of C_{60} / MAPbI_3 increased notably after adding I_3^- on top of the MAPbI_3 thin film (Fig. 3g), consistent with the tDOS results. Trap band I should therefore originate from iodide-related defects, probably iodide interstitials caused by the diffusion of I_3^- from the surface to the interior during the thermal annealing process because only iodide density was increased here. We then applied a positive bias of +1.2 V from PTAA to C_{60} . The positively charged bis(pyridine)iodine cations are too big to move, whereas I_i^- ions can drift under the electric field through more open material structures such as grain boundaries or amorphous regions. As shown in Fig. 3g, the accumulation locations of the defects for trap band I slightly drifted towards the PTAA side, validating the negative charge state of trap band I. Trap band I is also possibly caused by I_i^- in the form of I_3^- , which is too big to stay in

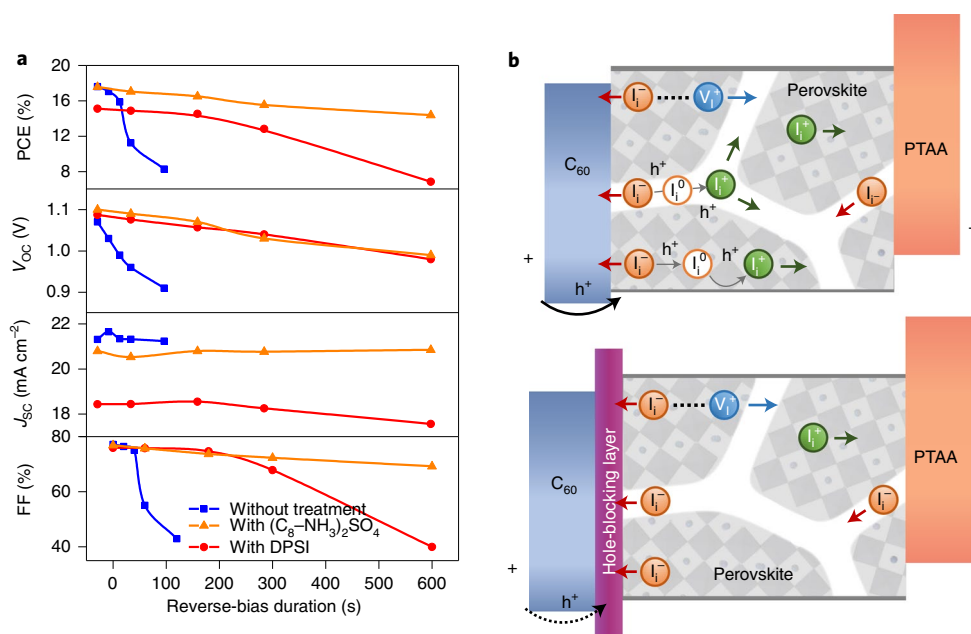


Fig. 4 | Improving the stability of perovskite solar cells under reverse bias. **a**, Changes in PCE, V_{oc} , short-circuit current density (J_{sc}) and FF with respect to the reverse-bias duration of the MAPbI₃ polycrystalline thin film solar cells without surface treatment and with (C₈-NH₃)₂SO₄ or DPSI treatment on the surface of MAPbI₃. **b**, Schematic illustrations of the degradation processes in MAPbI₃ polycrystalline thin film solar cells without and with a hole-injection blocking layer between C₆₀ and MAPbI₃. Without the hole-blocking layer, h⁺ can be readily injected from C₆₀ under reverse bias, as denoted by the solid black arrowed line. After adding a hole-blocking layer, hole injections are significantly retarded, as denoted by the dashed black arrowed line.

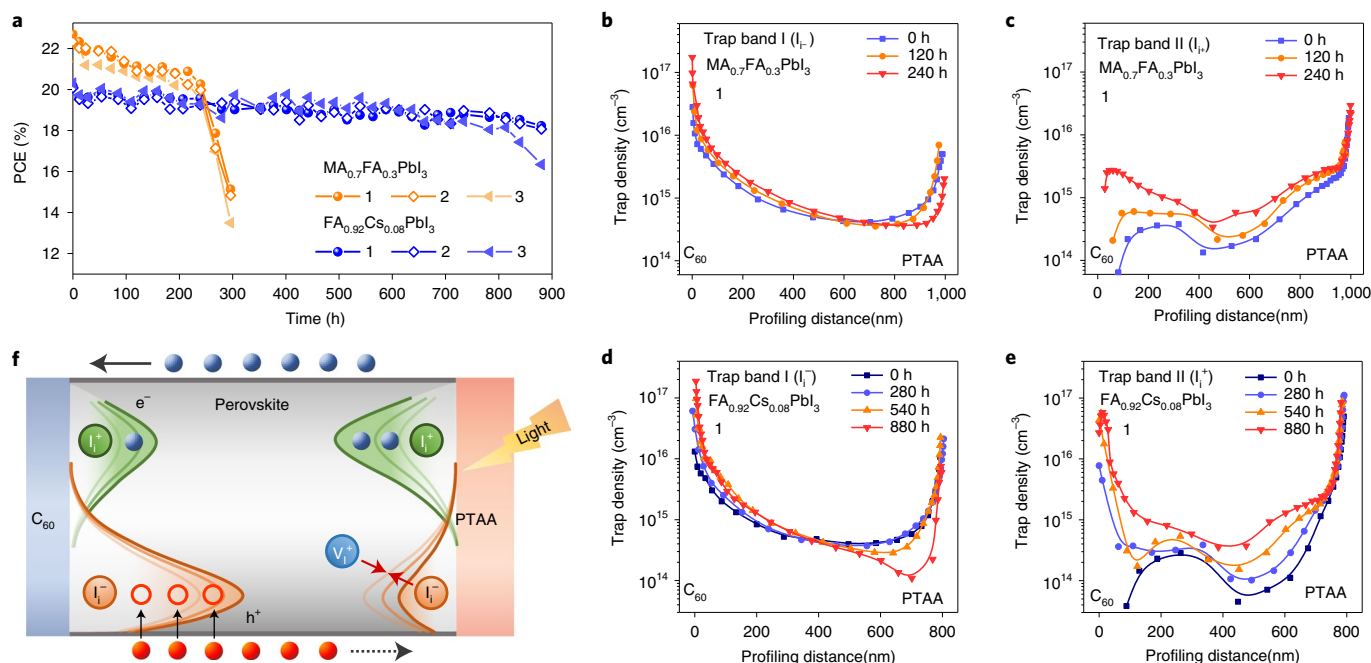


Fig. 5 | Degradation mechanism of perovskite solar cells under illumination. **a**, Statistical PCEs of p-i-n-structured MA_{0.7}FA_{0.3}PbI₃ and FA_{0.92}CS_{0.08}PbI₃ solar cells illuminated with one-sun light intensity at V_{oc} conditions for different periods of time. Three samples for each composition have been tested, denoted as 1, 2 and 3. **b, c**, Spatial distribution of the trap densities of trap band I (**b**) and trap band II (**c**) in the 1 MA_{0.7}FA_{0.3}PbI₃ solar cell after being illuminated at V_{oc} condition for different periods of time. **d, e**, Spatial distribution of the trap densities of trap band I (**d**) and trap band II (**e**) in the 1 FA_{0.92}CS_{0.08}PbI₃ solar cell after being illuminated at V_{oc} conditions for different periods of time. The note C₆₀ or PTAA indicates the locations that are close to the C₆₀ or PTAA layer of the device. The curves are guides to the eye. **f**, Schematic of the generation of iodide interstitial defects in a perovskite solar cell under illumination. The orange and green lines indicate the energetic and spatial distribution of degradation-generated I_i⁻ and I_i⁺ in the perovskite, respectively. The solid red and blue balls represent photogenerated holes and electrons, respectively. Photogenerated holes can be readily captured by I_i⁻ near the C₆₀ interface due to its short trapping time. The long-lived trapping state of electrons in I_i⁺ enables an efficient electron transport in the solar cell.

the bulk of grains but could abundantly exist at grain boundaries or amorphous regions of polycrystalline perovskite films³⁴. The DLCP results for trap band II are shown in Supplementary Fig. 13. With the increase of the forward biasing time, the trap density of trap band II gradually increased near the perovskite/PTAA interface and slightly expanded towards the interior of the perovskite film.

Therefore, the only two types of defect that can cause trap band II are I_i^+ or V_I^+ . Because I_i^+ and V_I^+ are unlikely to coexist in high concentrations, it is possible to find out which one is dominating trap band II by changing their concentrations. Here we changed these trap densities in MAPbI₃ polycrystalline thin films by varying PbI₂-to-MAI ratios. Previous calculations and experiments show that the formation of an iodide interstitial defect was favoured compared with iodide vacancies in a MAI-rich environment^{15,16,35}. As shown in Fig. 3b, the trap density of trap band II was remarkably increased in MAPbI₃ thin films synthesized with a MAI-rich precursor and was reduced using a PbI₂-rich precursor. The corresponding current density–voltage (*J*–*V*) curves of the solar cells are shown in Supplementary Fig. 14. Moreover, recent hybrid density functional theory calculations with better accuracy by considering spin–orbit coupling showed that V_I^+ does not cause deep trap states in MAPbI₃¹⁶. These results thus verify that trap band II is caused by I_i^+ , rather than V_I^+ .

In addition, we also investigated the change of traps in MA-free perovskites FA_{0.92}Cs_{0.08}PbI₃ and FA_{0.92}MA_{0.08}PbI₃. They showed similar variations of trap band II under reverse biases (Supplementary Note 2 and Supplementary Fig. 15), demonstrating the independence of trap band II on A site cations. Our recent work also indicated that trap band I in both MA_{0.7}FA_{0.3}PbI₃ and FA_{0.92}Cs_{0.08}PbI₃ was still caused by iodide interstitials^{36,37}, further verifying that A site cations did not cause deep traps in iodide perovskites, consistent with earlier theoretical predications². It is noted that this result does not exclude the migration of cations in perovskites, but it indicates the mobility of the cations is much smaller than halide-related defects, consistent with previous studies²⁹.

Reverse-bias degradation mechanism

After identifying the chemical nature of the defects for trap bands I and II in iodide perovskites, we can now explain the degradation mechanism of the solar cell under reverse bias. The MAPbI₃ device PCE substantially dropped from 17.4% to 8.3% after biasing at –1 V for 120 s (Fig. 2d and Supplementary Table 1), mostly caused by the decrease of the open-circuit voltage (*V*_{OC}) and fill factor (FF). The increase of charge traps mainly reduced the *V*_{OC} when the reverse-bias duration was short²⁴. With the extension of the reverse-bias duration, the FF started to drop in addition to the decrease of *V*_{OC}, which was probably due to the formation of conductive channels by the accumulation of mobile ions in perovskite^{38,39}. Moreover, it is found that the concentration and distribution of the doping and trap densities would roughly recover to their initial states when the solar cell was stored in the dark for about 2 h (Supplementary Fig. 10), and the solar cell performance recovered mostly—but not fully—to its initial state (Fig. 2d). Once the ions migrate away from their crystallographic positions under an electric field, they could not always completely recover to their initial statuses, thus ion migration may cause damages to perovskite crystal structure that is not recoverable.

To find out the origin of reverse bias-induced degradation, we first ruled out the chemical reactions between the Cu electrode and MAPbI₃ to be the cause of induced traps by changing the metal electrodes and interfacial layers (Supplementary Note 3 and Supplementary Fig. 16). Therefore, the reverse bias-induced charge traps are caused by the change of perovskites close to the electron transport layer (ETL). Under the reverse bias, excessive I_i^- is probably generated due to the formation of Frenkel defect pairs of I_i^-/V_I^+ via the interaction of mobile ions with the crystal lattice,

that is, perovskite $\rightarrow I_i^- + V_I^+$ (ref. 40,41). The negatively charged I_i^- moves towards the C₆₀/MAPbI₃ interface under the reverse bias, leading to the increase of trap band I there. At the C₆₀/MAPbI₃ interface, I_i^- first undergoes a trap-filling process with the injected holes (h^+) from the C₆₀ side, that is, $I_i^- + h^+ \rightarrow I_i^0$. I_i^0 would be further oxidized to I_i^+ followed by the valence band (VB) oxidation with the injected holes, that is, VB + $h^+ \rightarrow h^+@VB$, $h^+@VB + I_i^0 \rightarrow I_i^+$ (ref. 42), resulting in the accumulation of I_i^+ starting from the C₆₀/MAPbI₃ interface. The bimolecular reaction of $2I_i^0 \rightarrow I_2$ may also take place during the oxidation process, which leads to the loss of iodide from the perovskite and causes irreversible degradations. The above reactions involve sequential hole injections into the I_i^- trap states and the VB of the MAPbI₃, which should be closely related to the injection barrier for holes from the electrode to MAPbI₃.

To verify the proposed degradation mechanism, we treated the MAPbI₃ polycrystalline thin film with oxysalt of (C₈–NH₃)₂SO₄ or 3-(decyldimethylammonio)-propane-sulfonate inner salt (DPSI) and studied the stability of the solar cells under reverse bias. The wide-bandgap lead sulfate layer or the insulating DPSI layer was adopted to increase the potential barrier for the hole injection from the C₆₀ side⁹. Hole-only devices showed greatly reduced hole-injection currents from electrodes to perovskite after adding the lead sulfate or the DPSI layer (Supplementary Fig. 17). As a result, solar cells with the lead sulfate or DPSI layer between C₆₀ and MAPbI₃ showed significantly improved stabilities under reverse biases, as shown in Fig. 4a, Supplementary Fig. 18 and Supplementary Table 1. It is noted that lead sulfate or DPSI layers may also passivate the perovskite surface. To find out whether it is the hole-blocking effect of these interfacial layers that improves the reverse-bias stability, we fabricated perovskite solar cells with a structure of ITO/PTAA/perovskite/C₆₀/ALD-SnO₂/Cu, where ALD-SnO₂ is SnO₂ grown by atomic layer deposition. ALD-SnO₂ does not passivate perovskites due to the absence of direct contact, but it can reduce hole injections. The solar cells with the ALD-SnO₂ layer showed a significantly improved reverse-bias stability at the same reverse bias (Supplementary Fig. 19), confirming the role of hole blocking in enhancing device reverse-bias stability. Moreover, as seen in Supplementary Figs. 15 and 20, the perovskite solar cell with an ETL of SnO₂, which has a larger bandgap of ~3.8 eV and gives a larger potential barrier for hole injection, required a larger reverse bias of –3 V to trigger a similar reverse bias-induced efficiency reduction with the C₆₀–ETL-based solar cells¹³. These results well support the aforementioned mechanism for the degradation of the perovskite solar cells under reverse bias, which relies on the energy barrier for hole injections from the electrode to perovskites, as schematically illustrated in Fig. 4b.

Degradation under illumination

We further investigated the degradation of perovskite solar cells under illumination to understand the defect generation during solar cell operations. The change of PCEs over time for several typical p–i–n-structured MA_{0.7}FA_{0.3}PbI₃ and FA_{0.92}Cs_{0.08}PbI₃ solar cells under one-sun illumination at *V*_{OC} conditions is shown in Fig. 5a. The PCEs of MA_{0.7}FA_{0.3}PbI₃ solar cells gradually declined to ~90% of their initial efficiencies (T90) after 240 h illumination and then quickly dropped below 80% of their initial efficiencies (T80) in the following 30 h. For FA_{0.92}Cs_{0.08}PbI₃ solar cells, the PCE dropped by ~15% after illumination for about 900 h with a burn-in process at the beginning of the degradation. The burn-in process was ascribed to the migration of A site cations or the contact-layer interfaces⁴⁴, which could be recovered after leaving the solar cells in the dark⁴⁵. Here these cells were not intentionally optimized for best stability. We focused on the study of the degradation processes of the solar cells before T80, which is widely concerned for practical uses of solar cells.

We measured the changes of the carrier density distribution in these solar cells during degradation by DLCP (Supplementary Figs. 21 and 22) and derived the variations of trap distribution of I_i^- and I_i^+ in the solar cells. The distribution of I_i^- and I_i^+ in the $MA_{0.7}FA_{0.3}PbI_3$ and $FA_{0.92}Cs_{0.08}PbI_3$ solar cell after illuminating for different hours at V_{OC} conditions is shown in Fig. 5b–d. The results from more devices are shown in Supplementary Figs. 23 and 24 with better statistics. Their changes of the tDOS after degradation are shown in Supplementary Fig. 25. Interestingly, the trap density of I_i^+ gradually increased near both the C_{60} /perovskite and PTAA/perovskite interfaces during the degradation of both the $MA_{0.7}FA_{0.3}PbI_3$ and $FA_{0.92}Cs_{0.08}PbI_3$ solar cells, whereas that of I_i^- mainly enhanced near the C_{60} /perovskite interface but slightly decreased near the PTAA/perovskite interface. These results indicate that the initial degradation of the perovskite solar cells under illumination starts from the interface regions. Upon degradation, the C_{60} /perovskite interface became rich in both I_i^- and I_i^+ , potentially associated with the photo-induced decomposition of perovskite with the trapping of photogenerated electrons and holes by I_i^+ and I_i^- , respectively, followed by the degradation reactions of $2I_i^0 \rightarrow I_2$ and perovskite $\rightarrow I_i^- + V_i^+$ /perovskite $\rightarrow I_i^+ + V_i^+ + V_{Pb}^{2-}$ (ref. 18). The PTAA/perovskite interface showed an abundance of I_i^+ but a reduction of I_i^- . The opposite change of the defect densities of I_i^- and I_i^+ near the PTAA interface demonstrated the presence of a defect-annihilation process, which led to the decrease of the trap density related to I_i^- (ref. 18), as schematically shown in Fig. 5f. The defect-annihilation reaction requires sufficient migrations of iodide interstitials, whereas the perovskite close to the PTAA side was the most defective, which allowed faster ion migration. In addition, the higher photogenerated carrier density close to the PTAA interface further accelerates ion migrations. Therefore, the defect annihilation mainly occurred near the perovskite/PTAA interface where ion migration is the strongest^{24,46}.

To figure out which type of iodide interstitial was more detrimental to the solar cell efficiency, we used the photoluminescence and transient photocurrent decay (TPC) measurements to characterize the collection efficiency of electrons and holes individually in the degraded perovskite solar cells³⁷. The enhanced photoluminescence intensities in degraded solar cells at short-circuit conditions indicate that charge extractions from the perovskite layer became less efficient (Supplementary Fig. 26). To find out which type of the photogenerated carrier in degraded solar cells lags behind, we measured the carrier transit process in the solar cells before and after degradation with the laser coming in from C_{60} or PTAA sides. For both $MA_{0.7}FA_{0.3}PbI_3$ and $FA_{0.92}Cs_{0.08}PbI_3$ solar cells, the transport of holes were slowed down much more than that of electrons (Supplementary Fig. 26), indicating that holes are predominately trapped in degraded perovskites. This also validates that I_i^- as a hole-trapping defect is more detrimental to the perovskite solar cell efficiency than I_i^+ . This conclusion can be supported by previous studies that once the long-lived electron trapping induced by I_i^+ are filled by light-induced carriers upon initial illumination, they would not trap electrons in the following illumination. However, I_i^+ would still trigger the decomposition of perovskites given its long electron-trapping nature, leading to the decline of the solar cell efficiency indirectly¹⁸. In addition, we should note the generation of both I_i^- and I_i^+ are probably accompanied by V_i^+ , which does not introduce deep traps, but would provide pathways for ion migrations and phase segregations that eventually lead to the degradation of perovskite solar cells³⁷.

Conclusions

We identify the chemical nature of the dominant traps in iodide perovskites, which are detrimental to both the efficiency and the stability of their solar cells. It is found that iodide interstitials (I_i^- and I_i^+) dominate both the deep trap bands I (0.27 eV) and II (0.36 eV)

in $MAPbI_3$, as in $FA_xMA_{1-x}PbI_3$ and $FA_xCs_{1-x}PbI_3$ perovskites. Improved synthetic strategies to further reduce the iodide defects or passivate them in these perovskites are highly desired for the further improvement of the efficiency and stability of perovskite solar cells. Under reverse biases, $MAPbI_3$ solar cells start to degrade at the interface between ETL and $MAPbI_3$, whereas adding a blocking layer for charge injections at the interface could significantly suppress the degradation and improve the solar cell stability. Under illumination and V_{OC} conditions, perovskite solar cells begin to degrade at the interfacial regions between perovskites and charge transport layers, whereas the efficiency of the solar cell is mainly limited by the I_i^- defects at the perovskite/ C_{60} interface as defect annihilation of iodide interstitials and vacancies mainly occurs at the perovskite/PTAA interface. Further improvement on the stability of perovskite solar cells should be addressed on suppressing the generation of iodide interstitial defects close to the interfaces.

Methods

Materials. Unless stated otherwise, all the materials and solvent were purchased from Sigma-Aldrich. Lead iodide (PbI_2) with a purity of 99.995% was purchased from BeanTown Chemical. MAI was purchased from GreatCell Solar.

Synthesis of perovskite thin single crystals. The precursor solution for the synthesis of $MAPbI_3$ thin single crystals was prepared by dissolving MAI and PbI_2 with the molar ratio of 1:1 in γ -butyrolactone at a concentration of 1.5 M. The solution was heated at 65 °C with magnetic stirring until the solutes fully dissolved, followed by filtering with a 0.2 μ m pore size PTFE filter. The $MAPbI_3$ thin single crystals were grown by a hydrophobic interface-confined lateral crystal-growth method. A PTAA solution with the concentration of 2 mg ml⁻¹ in toluene was first spin coated onto an ITO glass substrate at 4,000 r.p.m. for 30 s. The prepared PTAA film was thermally annealed at 100 °C for 10 min. Then, a drop of $MAPbI_3$ solution was inserted into two PTAA/ITO substrates, which were then placed on a hot plate at a temperature of 80 °C for 1 h. The temperature was subsequently increased to 85 °C and 90 °C to further promote the growth of the thin single crystals. Finally, the two substrates were separated to obtain $MAPbI_3$ thin single crystals on PTAA/ITO substrates. The thickness of the thin single crystal was about 35 μ m.

Fabrication of perovskite solar cells. For the fabrication of the $MAPbI_3$ polycrystalline thin film solar cell, ITO glass substrates were first prepared by cleaning with detergent, deionized water, acetone and isopropanol in sequence and ultraviolet ozone treatment (20 min). A PTAA solution with the concentration of 2 mg ml⁻¹ in toluene was spin coated onto the ITO glass substrate at 4,000 r.p.m. for 30 s. The prepared PTAA film (10 nm) was thermally annealed at 100 °C for 10 min. An 80 μ l 1.3 M $MAPbI_3$ precursor solution in dimethylformamide and dimethyl sulfoxide (with volume ratio of 9:1) was spin coated onto the PTAA at 2,000 r.p.m. for 2 s and then 4,000 r.p.m. for 20 s. During the spin coating, 130 μ l of toluene was added into the film at 8 s. The formed film was then heated at 65 °C for 10 min and 100 °C for 10 min. The $FA_{0.92}Cs_{0.08}PbI_3$ perovskite films were fabricated by blade coating at room temperature under nitrogen knife blowing by using a precursor solution containing 1.0 M $FAPbI_3$ and 0.09 M $CsPbI_3$ dissolved in a 2-methoxyethanol:dimethyl sulfoxide solvent mixture. The coated solid film was annealed at 150 °C in air for 2 mins. For the fabrication of the $MA_{0.7}FA_{0.3}PbI_3$ perovskite, precursor solutions (2.5 M $MAPbI_3$ and 1.67 M $FAPbI_3$) were prepared separately by dissolving corresponding organic halides and lead iodide in 2-mercaptoethanol and stored in a N_2 -filled glovebox. Before blade coating, the $MAPbI_3$ and $FAPbI_3$ precursor solutions were mixed and diluted to a 1.37 M $MA_{0.7}FA_{0.3}PbI_3$ solution. Subsequently, the precursor solution was blade coated onto the PTAA-covered ITO glass substrates with a gap of 250 μ m at a movement speed of 20 mm s⁻¹. The N_2 knife worked at 20 psi during blade coating. After that, the perovskite films were annealed at 120 °C for 5 min in air. The whole device was completed by sequentially thermal evaporating C_{60} (30 nm), BCP (6 nm) and Cu electrodes (80 nm) on the perovskite film. For the fabrication of the $FA_{0.92}MA_{0.08}PbI_3$ n-i-p-structure solar cell, the SnO_2 layer was deposited by spin coating a SnO_2 nanoparticle solution in H_2O with a concentration of 2.67% onto the ITO at 4,000 r.p.m. for 30 s, followed by thermal annealing in ambient air at 150 °C for 30 min. After cleaning the SnO_2 layer with ultraviolet ozone for 10 min to improve the surface wetting, 1.5 M of PbI_2 in dimethylformamide:dimethyl sulfoxide (9:1) was spin coated onto SnO_2 at 1,500 r.p.m. for 30 s. The film was then annealed at 70 °C for 1 min. After cooling down to room temperature, a solution of FAI:MAI:MACI (90 mg:6.39 mg:9 mg in 1 ml isopropanol) was spin coated onto the PbI_2 at 2,000 r.p.m. for 30 s, followed by thermal annealing at 150 °C for 15 min in ambient air conditions (30–40% humidity). After the formation of the perovskite layer, the sample was transferred to a nitrogen-atmosphere glovebox for the deposition of Spiro-OMeTAD. The Spiro-OMeTAD solution (72.3 mg Spiro-OMeTAD powder, 35 μ l bis(trifluoromethane) sulfonamide lithium salt (Li-TFSI) solution (260 mg ml⁻¹ stock solution in acetonitrile),

30 μl 4-tertbutylpyridine, all in 1 ml chlorobenzene) was spin coated at 2,000 r.p.m. for 30 s. Finally, Ag was thermally evaporated on the Spiro-OMeTAD to form the 120 nm-thick electrodes with a shadow mask of 0.8 cm^2 . The thickness of the MAPbI_3 , $\text{FA}_{0.92}\text{Cs}_{0.08}\text{PbI}_3$, $\text{MA}_{0.7}\text{FA}_{0.3}\text{PbI}_3$ and $\text{FA}_{0.92}\text{MA}_{0.08}\text{PbI}_3$ thin films were about 500 nm, 900 nm, 1,000 nm and 800 nm, respectively, which were measured by a Bruker Dektak XT stylus profiler. The MAPbI_3 thin single-crystal solar cell was fabricated by sequentially thermal evaporating C_{60} (30 nm), BCP (6 nm) and Cu electrodes (80 nm) on the perovskite crystal that was grown on the PTAA/ITO substrates. For the incorporation of I_3^- onto the surface region of MAPbI_3 polycrystalline thin films and thin single crystals, 0.5 mg ml^{-1} I_2 in isopropyl alcohol and 0.2 mg ml^{-1} P4VP in isopropyl alcohol were first mixed with a volume ratio of 1:1, followed by dynamic spin coating of the solution on the surface of the MAPbI_3 thin film and single crystal at 5,000 r.p.m. for 30 s. The film and single crystal were then heated at 65 $^\circ\text{C}$. The oxy-salt treatment of the MAPbI_3 polycrystalline thin film was carried out by spin coating 2 mM $(\text{C}_6\text{-NH}_2)_2\text{SO}_4$ solution in a toluene/isopropyl alcohol mixture on the fabricated MAPbI_3 polycrystalline thin film at 5,000 r.p.m. for 30 s, followed by thermal annealing at 100 $^\circ\text{C}$ for 10 min. The DPSI layer was formed by dynamic spin coating of 0.5 mg ml^{-1} DPSI solution in isopropyl alcohol on the fabricated MAPbI_3 polycrystalline thin film at 5,000 r.p.m. for 30 s. The ALD- SnO_x layer was prepared with the Ultratech/Cambridge Nanotech Savannah S200 system at a base pressure of 0.25 torr. Tetrakis(dimethylamino)tin(IV) (60 $^\circ\text{C}$) and H_2O (25 $^\circ\text{C}$) were used as precursors for SnO_x fabrication. The dosing sequences (t_1 , t_2 , t_3 , t_4), where t_1 and t_3 refer to the times of precursor and oxidant dosing, and t_2 and t_4 refer to the times for purging, are (0.35 s, 15 s, 0.25 s, 15 s). The temperature of the chamber was varied from 80 $^\circ\text{C}$ to 100 $^\circ\text{C}$. The growth rate of the ALD- SnO_x layer at 100 $^\circ\text{C}$ is ~ 1.6 Å per cycle. All the solar cells were encapsulated in glass with the edges being sealed by A-B glue (epoxy) after fabrication. The A-B glue (epoxy) was premixed before use and aged for 12 h before further handling.

Electrical measurements. The J - V measurement was performed under the simulated air mass 1.5 spectra (AM 1.5 G) irradiation (100 mW cm^{-2}) produced by a xenon lamp-based solar simulator (Oriol 94043A, 450 W AAA Solar Simulator). The light intensity was calibrated by a reference solar cell and meter (91150, Newport). A Keithley 2400 source meter was used for the J - V measurements. All J - V measurements were taken under 100 mW cm^{-2} illumination at reverse-scan mode with a scan rate of 0.05 V s^{-1} . For the TPC measurement, solar cells were connected to an oscilloscope (DOS-X 3104A) with an internal impedance of 50 Ω to form short-circuit conditions. An attenuated 337 nm pulsed laser (SRS NL 100 Nitrogen Laser, frequency of 10 Hz and pulse width of <3.5 ns) was used from either the C_{60} or PTAA side of the device. The corresponding photocurrent decay was recorded by the oscilloscope. The RC constant of the circuit was around 100 ns. For the degradation study of perovskite solar cells, the efficiencies of more than nine devices from three batches for each composition were tracked, and three devices for each composition, which represented the average degradation rate of these solar cells under illumination, were chosen for the TAS and DLCP measurements.

Capacitance measurements. The TAS and DLCP measurements were performed using an Agilent E4980A precision LCR meter. For the TAS measurement, the d.c. bias (V) was fixed at 0 V, and the amplitude of the a.c. bias (δV) was 20 mV. The scanning range of the a.c. frequency (f) was 0.02–2,000 kHz. The tDOS ($N_T(E_\omega)$) is calculated using the equation $N_T(E_\omega) = -\frac{1}{qkT} \frac{\partial dC}{\partial \omega} \frac{V_\omega}{W}$, where q , k , T , ω and C are elementary charge, Boltzmann's constant, temperature, angular frequency and specific capacitance, respectively. W and V_ω are the depletion width and build-in potential, respectively. The demarcation energy, $E_\omega = kT \ln(\frac{\omega_0}{\omega})$, where ω_0 is the attempt-to-escape angular frequency that equals to $2\pi\nu_0 T^2$, is derived from the temperature-dependent C - f measurements. The reduced attempt-to-escape frequency ν_0 is obtained from the fitting of $\ln(\frac{f}{\omega}) = \frac{E_\omega}{kT} - \ln(2\pi\nu_0)$ obtained at different T . In this work, the ω_0 for trap bands I and II were individually derived from the temperature-dependent C - f measurements. Each trap band was plotted with a Gaussian distribution in energy and its own ω_0 to determine the trap depth from the band edge. We should note that the TAS method cannot distinguish between states close to the valence band or the conduction band. Temperature-dependent capacitance measurements were carried out in a Lake Shore Cryotronics probe stage with a Lake Shore Cryotronics temperature controller model 336. SMA connected cables and ZN50R d.c./radio frequency probes that were applicable for radio frequency tests up to 1 GHz were equipped with the probe station. Before each measurement, the system was self-calibrated under open-circuit and short-circuit conditions to compensate for any undesired signal from the instrument. For the DLCP measurements, the V was scanning from 0 V to the V_{oc} (for example, 1.1 V) for the perovskite solar cells. The DLCP method uses a series of variable δV (for example, 20–200 mV) to measure the junction capacitance and acquire the capacitance contribution from the trap states by taking advantage of the information embedded in the higher-order terms. The capacitance measured at each δV was recorded and fitted with a polynomial function $C = C_0 + C_1\delta V + C_2(\delta V)^2 + \dots$ to obtain C_0 and C_1 . With the determination of C_0 and C_1 , the total carrier density (N) that includes both free carrier density and trap density at the profiling distance X from the junction barrier is calculated by $N = -\frac{C_0}{2q\epsilon A^2 C_1}$, where q is the elementary charge, ϵ is the dielectric constant

of the semiconductor (which is 31, 33 and 33 for MAPbI_3 , $\text{FA}_{0.92}\text{Cs}_{0.08}\text{PbI}_3$ and $\text{MA}_{0.7}\text{FA}_{0.3}\text{PbI}_3$, respectively) and A is the active area of the junction. The profiling distance from the junction barrier was calculated by $\frac{\epsilon A}{C_0}$, which was changed by tuning the V . For each a.c. bias, an additional offset d.c. voltage was applied to keep the maximum forward bias constant. The trap density within a certain trap depth range was calculated by subtracting the total carrier density measured at a larger E_ω (lower a.c. frequency) with that measured at a smaller E_ω (higher a.c. frequency).

XPS measurement. The XPS was measured by a Kratos Axis Ultra DLD X-ray Photoelectron Spectrometer with an X-ray source providing monochromatic Al-K α source.

Photoluminescence measurement. The photoluminescence measurement was carried out with a PicoQuant FT300 spectrometer. A 640 nm pulsed laser with a repetition rate of 40 MHz with an intensity of ~ 0.9 sun per pulse was adopted for the measurement.

Reporting Summary. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data availability

All data generated or analysed during this study are included in the article, its Supplementary Information and its data source. Source data are provided with this paper.

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Author contributions

J.H. and Z.N. conceived the idea and designed the experiments. Z.N., H.J., C.F., H.G., S.X., G.Y. and Q.J. fabricated the perovskite polycrystalline thin film solar cells. Z.N. and Y.L. synthesized the perovskite thin single crystals. Z.N. and S.X. carried out the measurements for the devices. Z.Y. performed the ALD for the devices. Y.Y. participated in data analyses and project discussion. Y.D. did the TPC measurement. Z.N. and J.H. wrote the Article, and all authors reviewed it.

Competing interests

The authors declare no competing interests.

Additional information

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► Experimental design

Please check: are the following details reported in the manuscript?

1. Dimensions

Area of the tested solar cells	☒ Yes ☐ No	Section "Method", Fabrication of perovskite solar cells.
Method used to determine the device area	☐ Yes ☒ No	The area of 0.8 cm ² is determined by the designed shadow mask and easy to be measured by a caliper.

2. Current-voltage characterization

Current density-voltage (J-V) plots in both forward and backward direction	☒ Yes ☐ No	Supplementary Figure 4
Voltage scan conditions <i>For instance: scan direction, speed, dwell times</i>	☒ Yes ☐ No	Section "Method", Electrical measurements.
Test environment <i>For instance: characterization temperature, in air or in glove box</i>	☒ Yes ☐ No	Section "Method", Electrical measurements & Fig. 2, Fig. 5 and Supplementary Figure 15
Protocol for preconditioning of the device before its characterization	☐ Yes ☒ No	No preconditioning processes for the solar cells
Stability of the J-V characteristic <i>Verified with time evolution of the maximum power point or with the photocurrent at maximum power point; see ref. 7 for details.</i>	☐ Yes ☒ No	The light stability is conducted under the Voc degradation condition

3. Hysteresis or any other unusual behaviour

Description of the unusual behaviour observed during the characterization	☒ Yes ☐ No	Sections "Chemical nature of defects" and "Degradation under illumination".
Related experimental data	☒ Yes ☐ No	Fig. 2, Fig. 5 and Supplementary Figure 15

4. Efficiency

External quantum efficiency (EQE) or incident photons to current efficiency (IPCE)	☐ Yes ☒ No	Not relevant to the key study of the chemical nature of defects of the manuscript.
A comparison between the integrated response under the standard reference spectrum and the response measure under the simulator	☐ Yes ☒ No	Not relevant to the key study of the chemical nature of defects of the manuscript.
For tandem solar cells, the bias illumination and bias voltage used for each subcell	☐ Yes ☒ No	No tandem solar cells involved.

5. Calibration

Light source and reference cell or sensor used for the characterization	☒ Yes ☐ No	Section "Method", Electrical measurements.
Confirmation that the reference cell was calibrated and certified	☒ Yes ☐ No	Section "Method", Electrical measurements.

Calculation of spectral mismatch between the reference cell and the devices under test

☐ Yes
☒ No

Calculation is not reported but we reported all needed information including our device spectrum and the model of Silicon reference cell (Newport 91150V-KG5) in Section "Method", Electrical measurements.

6. Mask/aperture

Size of the mask/aperture used during testing

☐ Yes
☒ No

No mask used during the test.

Variation of the measured short-circuit current density with the mask/aperture area

☐ Yes
☒ No

No mask used during the test.

7. Performance certification

Identity of the independent certification laboratory that confirmed the photovoltaic performance

☐ Yes
☒ No

Not relevant to the key study of the chemical nature of defects of the manuscript.

A copy of any certificate(s)

Provide in Supplementary Information

☐ Yes
☒ No

Not relevant to the key study of the chemical nature of defects of the manuscript.

8. Statistics

Number of solar cells tested

☒ Yes
☐ No

Section "Method", Fabrication of perovskite solar cells and electrical measurement. Perovskites with different compositions are used for the statistic analysis. For the degradation study of perovskite solar cells, the efficiencies of more than 9 devices from 3 batches for each composition were tracked, and 3 devices for each composition which represented the average degradation rate of these solar cells under illumination were chosen for the TAS and DLCP measurements.

Statistical analysis of the device performance

☒ Yes
☐ No

Figs. 1, 2, 5 and Supplementary Figure 3, 15, 23, 24, 25.

9. Long-term stability analysis

Type of analysis, bias conditions and environmental conditions

For instance: illumination type, temperature, atmosphere

humidity, encapsulation method, preconditioning temperature

☒ Yes
☐ No

Section "Method", Electrical measurements & Fig. 5.